Efficacy of proton pump inhibitor (PPI) therapy in treating sleep disturbances in patients with gastroesopahgeal reflux disease (GERD): a systematic review and meta-analysis

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Review question

What is the clinical effectiveness of proton pump inhibitor (PPI) therapy on sleep disturbances in patients with gastroesophageal reflux disease (GERD)?

Searches

Electronic search
The following databases should be used
1. the Cochrane Central Register of Controlled Trials (CENTRAL);
2. MEDLINE via PubMed;
3. EMBASE;
See Appendix 1, 2, and 3 for the search strategies.

Other resources
The following sites should also be used
1. the World Health Organization International Clinical Trials Platform Search Portal (ICTRP)
See Appendix 4 for the search strategies.
2. ClinicalTrials.gov
See Appendix 5 for the search strategies.

We will also check references for extracted studies and international guidelines. We will contact the original authors for any unclear data.

Types of study to be included

Prospective, randomized controlled trials describing PPI therapy for patients with GERD and sleep disturbances will be assessed for eligibility. Crossover studies, cluster randomized studies, will be excluded. Studies conducted by quasi-experimental methods will be excluded. Retrospective studies will be excluded. No exclusion will be made for observation periods.

Condition or domain being studied

Symptoms of GERD and sleep during nighttime.

Participants/population

Inclusion criteria
• Adults (>18 years of age) with symptoms of gastroesophageal reflux disease during nighttime.

• Prospective, controlled articles describing treatments for nocturnal GERD will be assessed for eligibility.

Exclusion criteria:
• Prospective non-controlled, retrospective articles, case series, and case reports.
• Studies published before 1990, as PPIs entered the market in the late 1980s.

Intervention(s), exposure(s)
PPI therapy

Comparator(s)/control

Placebo or control group.

Main outcome(s)

Change in sleep time and quality

**** Measures of effect***
Change in polysomnography measures (such as apnea-hypopnea index (AHI))
Change in non-polysomnography measures (such as the patient sleep quality index (PSQI), work productivity and activity index (WPAI), Epworth sleepiness scale (ESS), and other health-related quality of life [HRQOL] questionnaires)

Additional outcome(s)

Change in pH impedance parameters (acid exposure, number of refluxes episodes)
Change in reflux symptoms
All adverse events: Proportion of people who developed an adverse event according to the original author's definition

**** Measures of effect***
Improvement in subjective and objective sleep parameters, changes in esophageal pH parameters, number of reflux episodes or improvement in nocturnal GERD symptoms.

Data extraction (selection and coding)

Two authors will independently screen the titles of all references retrieved by the literature search. Irrelevant studies and duplicates will be removed. The remaining citations will be retrieved in full text and screened independently by the same two reviewers. If a study is abstract only and it is not clear whether it meets the criteria for review, the authors will contact the original author. Disagreements between two reviewers should be discussed and resolved. If necessary, the authors will discuss with a third reviewer.

Risk of bias (quality) assessment

The Cochrane risk of bias tool 2 will be used. Two reviewers will perform the quality assessment. A third author will resolve disagreements between judgements.

Strategy for data synthesis

We will meta-analyze the standardized mean difference (SMD) and 95% confidence interval with the following continuous variables: questionnaire responses, pH-impedance results, polysomnography results
We will follow the method of the Cochrane Handbook[1] for the integration of the mean and standard deviation of continuous variables.

Dealing with missing values

We will not complete the data of dropouts according to the recommendations of the Cochrane Handbook[1]. We will perform a meta-analysis of the data presented by the original authors. We will contact the original author about missing values. If only the standard error or p-value is reported, we will calculate the standard deviation using the method of Altman[2]. If the authors could not provide the missing data, the standard deviation should be calculated from the confidence interval and t-value using the method described in the Cochrane Handbook[1]. Alternatively, a validated method[3] should be used to supplement it. The validity of these methods will be verified by sensitivity analysis.

Analysis of heterogeneity

We will evaluate heterogeneity visually using a forest plot. Simultaneously, we will calculate the I2 values (I2 values of 0% to 40%: might not be important; 30%to 60%: may represent moderate heterogeneity; 50% to 90%: may represent substantial heterogeneity; 75% to 100%: considerable heterogeneity). If heterogeneity is detected (I2 > 50%), the cause of the heterogeneity will be examined; Cochrane Chi2 test (Q-test) will be performed for I2 values, and a P value <0.10 will be considered statistically significant.

Evaluation of publication bias

We will search clinical trial registration sites (ClinicalTrials.gov, ICTRP) for studies that have been completed but not yet published. At the same time, we will visually assess the studies using funnel plots and perform the Egger test; a P value <0.10 will be considered statistically significant. If we find only 10 or fewer studies, or if we find only studies with similar sample sizes, we will not perform the study.

Subgroup analysis, search for heterogeneity

The following subgroup analyses will be conducted to assess the heterogeneity of clinical study participants and interventions.

1.(For P) Age, sex, BMI, smoking status, alcohol use, presence of psychiatric comorbidities, socioeconomic status, shift worker status, heart failure

2. (For I) PPI dose, PPI type

Sensitivity Analysis

The following sensitivity analyses will be conducted for the primary outcome:
1. Exclusion of non-double-blind studies.
2. Random-effects instead of a fixed-effect model.
3. Exclusion of studies using imputed statistics.
4. Missing participants:

-Best-best scenario: all missing patients in the two groups remain unchanged

-Best-worst scenario: all missing patients in the intervention group remain unchanged and all missing patients in the control group have outcomes

-Worst-best scenario: all missing patients in the intervention group have outcomes and all missing patients in the control group remain unchanged

Primary analysis (worst-worst scenario: all missing patients in the two groups have outcomes)

Summary of findings (SOF) table

We will create SOF tables for the following outcomes based on the recommendations of the Cochrane Handbook[1]:

1. Change in polysomnography measures (such as apnea-hypopnea index (AHI))
2. Change in non-polysomnography measures (such as the patient sleep quality index (PSQI), work productivity and activity index (WPAI), Epworth sleepiness scale (ESS), and other health-related quality of life [HRQOL] questionnaires)
3. Change in pH impedance parameters (acid exposure, number of refluxes episodes)
4. Change in reflux symptoms
5. All adverse events: Proportion of people who developed an adverse event according to the original author's definition

Each SOF includes a grading of the evidence based on the GRADE approach[1,5].

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Type and method of review

Intervention, Systematic review and Meta-analysis

Anticipated or actual start date

01 March 2021

Anticipated completion date

01 March 2022

Funding sources/sponsors

None

Grant number(s)
State the funder, grant or award number and the date of award
None

Conflicts of interest

Language

English

Country

United States

Stage of review

Review Ongoing

Subject index terms status

Acid reflux, heartburn, nocturnal reflux, gastric acid reflux disease, esophageal reflux, gastro-oesophageal reflux disease

Subject index terms

MeSH headings
- MeSH heading term for GERD is “Gastroesophageal Reflux”
- MeSH heading term for PPI is : “Proton Pump Inhibitor” “Omeprazole” “Esomeprazole” “Lansoprazole” “Pantoprazole” “Dexlansoprazole” “Rabeprazole”

Appendix 1: CENTRAL search strategy
([mh "gastroesophageal reflux"] OR GERD:ti,ab OR GORD:ti,ab OR (([mh "gastroesophageal reflux"] OR (gastroesophageal AND reflux) OR "gastroesophageal reflux") AND "acid reflux":ti,ab) OR "acid reflux":ti,ab OR heartburn:ti,ab OR "nocturnal reflux":ti,ab OR "gastric acid reflux disease":ti,ab OR "esophageal reflux":ti,ab OR "gastro esophageal reflux disease":ti,ab OR "gastroesophageal reflux disease":ti,ab) AND ([mh "proton pump inhibitors"] OR ([mh omeprazole] OR [mh esomeprazole]) OR [mh esomeprazole] OR [mh lansoprazole] OR [mh pantoprazole] OR [mh dexlansoprazole] OR [mh rabeprazole] OR "proton pump inhibitor":ti,ab OR omeprazole:ti,ab OR esomeprazole:ti,ab OR lansoprazole:ti,ab) AND ((
"randomized controlled trial[Publication Type]"
OR
"controlled clinical trial[Publication Type]"
OR randomized:ti,ab OR
"drug therapy[MeSH Subheading]"
OR placebo:ti,ab OR randomly:ti,ab OR trial:ti,ab OR groups:ti,ab) NOT ([mh animals] NOT [mh humans]))

Appendix 2: MEDLINE (via PubMed) search strategy
("gastroesophageal reflux"[MeSH Terms] OR "GERD"[Title/Abstract] OR "GORD"[Title/Abstract] OR (("gastroesophageal reflux"[MeSH Terms] OR ("gastroesophageal"[All Fields] AND "reflux"[All Fields]) OR "gastroesophageal reflux"[All Fields]) AND "acid reflux"[Title/Abstract]) OR "acid reflux"[Title/Abstract] OR "heartburn"[Title/Abstract] OR "nocturnal reflux"[Title/Abstract] OR "gastric acid reflux disease"[Title/Abstract] OR "esophageal reflux"[Title/Abstract] OR "gastro esophageal reflux disease"[Title/Abstract] OR "gastroesophageal reflux disease"[Title/Abstract]) AND ("proton pump inhibitors"[MeSH Terms] OR ("omeprazole"[MeSH Terms] OR "esomeprazole"[MeSH Terms]) OR "esomeprazole"[MeSH Terms] OR "lansoprazole"[MeSH Terms] OR "pantoprazole"[MeSH Terms] OR "dexlansoprazole"[MeSH Terms] OR "rabeprazole"[MeSH Terms] OR "proton pump inhibitor"[Title/Abstract] OR "omeprazole"[Title/Abstract] OR "esomeprazole"[Title/Abstract] OR "lansoprazole"[Title/Abstract]) AND (("randomized controlled trial"[Publication Type] OR "controlled clinical trial"[Publication Type] OR "randomized"[Title/Abstract] OR "drug therapy"[MeSH Subheading] OR "placebo"[Title/Abstract] OR "randomly"[Title/Abstract] OR "trial"[Title/Abstract] OR "groups"[Title/Abstract]) NOT ("animals"[MeSH Terms] NOT "humans"[MeSH Terms]))

Appendix 3: EMBASE search strategy
('gastroesophageal reflux'/exp OR GERD:ti,ab OR GORD:ti,ab OR (('gastroesophageal reflux'/exp OR (gastroesophageal AND reflux) OR "gastroesophageal reflux") AND "acid reflux":ti,ab) OR "acid reflux":ti,ab OR heartburn:ti,ab OR "nocturnal reflux":ti,ab OR "gastric acid reflux disease":ti,ab OR "esophageal reflux":ti,ab OR "gastro esophageal reflux disease":ti,ab OR "gastroesophageal reflux disease":ti,ab) AND ('proton pump inhibitors'/exp OR ('omeprazole'/exp OR 'esomeprazole'/exp) OR 'esomeprazole'/exp OR 'lansoprazole'/exp OR

'pantoprazole'/exp OR 'dexlansoprazole'/exp OR 'rabeprazole'/exp OR "proton pump inhibitor":ti,ab OR omeprazole:ti,ab OR esomeprazole:ti,ab OR lansoprazole:ti,ab) AND ((
"randomized controlled trial[Publication Type]"
OR
"controlled clinical trial[Publication Type]"
OR randomized:ti,ab OR
"drug therapy[MeSH Subheading]"
OR placebo:ti,ab OR randomly:ti,ab OR trial:ti,ab OR groups:ti,ab) NOT ('animals'/exp NOT 'humans'/exp))

Appendix 4: ICTRP search strategy

Participants: GERD OR GORD OR Gastroesophageal Reflux OR Acid reflux OR heartburn OR nocturnal reflux OR gastric acid reflux disease OR esophageal reflux OR gastro-esophageal reflux disease

Intervention: Proton pump inhibitor OR Omeprazole OR Esomeprazole OR Lansoprazole OR Pantoprazole OR Dexlansoprazole OR Rabeprazole

Appendix 5: ClinicalTrials.gov search strategy

Participants: GERD OR GORD OR Gastroesophageal Reflux OR Acid reflux OR heartburn OR nocturnal reflux OR gastric acid reflux disease OR esophageal reflux OR gastro-esophageal reflux disease

Intervention: Proton pump inhibitor OR Omeprazole OR Esomeprazole OR Lansoprazole OR Pantoprazole OR Dexlansoprazole OR Rabeprazole

Appendix 6 Reference

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